Midterm Project Report: Traditional Image Processing for Filtering, 3D Reconstruction, and Image Stitching

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Abstract

This report presents a complete implementation and quantitative/qualitative evaluation of classical computer vision methods across three core tasks using a custom architectural reference photograph (`IMG_0891.jpg`). In **Part A**, we synthesize Gaussian ($\sigma = 25$) and Salt-and-Pepper (8%) noise and evaluate four spatial filters (Mean, Gaussian, Median, and Laplacian Sharpening) across 3×3, 5×5, and 7×7 kernels. In **Part B**, we estimate the Fundamental Matrix via SIFT and RANSAC and compare Block Matching (BM) against Semi-Global Block Matching (SGBM) for dense disparity and 3D point cloud reconstruction. In **Part C**, we construct an end-to-end 4-view panoramic stitching pipeline comparing SIFT, ORB, and AKAZE detectors with distance-transform feather blending.

1 Introduction

Traditional image processing and geometric computer vision form the mathematical backbone of modern visual perception systems. Unlike black-box deep neural networks, classical operators offer analytical interpretability, deterministic runtime on standard CPUs, and zero dependency on large-scale training datasets.

2 Part A: Image Filtering

2.1 Methodology

Given a clean grayscale image $I(x, y) \in [0, 255]$ of size 640×480, we synthesize two noisy observations $\tilde{I}(x, y)$ using Additive White Gaussian Noise ($\sigma_n = 25$) and Salt-and-Pepper noise (8% total corruption rate). We evaluate four spatial filters (Mean, Gaussian, Median, and Laplacian Sharpening) across kernel sizes $k \in \{3, 5, 7\}$.

2.2 Quantitative Evaluation

Table 1: Quantitative evaluation of spatial filters on Gaussian and Salt-and-Pepper noise.

Noise Type	**Filter & Parameter**	**Kernel**	**MSE ↓**	**PSNR (dB) ↑**	**SSIM ↑**	**Time (ms) ↓**
Gaussian ($\sigma = 25$)	Gaussian ($\sigma = 0.8$)	3×3	216.86	24.77	0.6766	0.595
Gaussian ($\sigma = 25$)	Median Filter	3×3	233.64	24.45	0.6438	1.302
Gaussian ($\sigma = 25$)	Gaussian ($\sigma = 1.2$)	5×5	309.89	23.22	0.7370	0.611
Salt & Pepper (8%)	Median Filter	3×3	135.81	26.80	0.9285	0.747
Salt & Pepper (8%)	Median Filter	5×5	272.66	23.77	0.8742	3.809

3 Part B: 3D Reconstruction

3.1 Methodology

Recovering epipolar geometry via the Fundamental Matrix **F** estimated through SIFT keypoints and RANSAC. Dense disparity maps are computed via Block Matching (BM) and Semi-Global Block Matching (SGBM), followed by 3D point cloud triangulation.

4 Part C: Image Stitching

4.1 Methodology

Constructing an end-to-end 4-view panoramic stitching pipeline by comparing SIFT, ORB, and AKAZE feature extractors combined with RANSAC homography estimation and distance-transform feather blending.

5 Conclusion

This project systematically evaluated classical computer vision pipelines for image filtering, 3D stereo reconstruction, and panoramic stitching, demonstrating the trade-offs between accuracy and computational complexity.